

Adult Income Data Cleaning and Exploratory Data Analysis

Descriptive analysis of the UCI Adult Income Dataset

1. Project Summary

This project downloads, cleans, and analyzes the UCI Adult Income dataset. The analysis combines the official `adult.data` and `adult.test` files and examines demographic, education, employment, and working-hour characteristics associated with annual income above USD 50K.

The analysis is descriptive. The observed relationships represent associations in the historical dataset and should not be interpreted as evidence that a particular characteristic causes a person's income.

2. Problem Statement

Main question: Which demographic, education, and work characteristics are associated with an annual income above USD 50K in this census dataset?

3. Dataset Information

The combined dataset contains 48,842 records and 20 analysis columns after cleaning and feature creation.

The dataset contains numerical and categorical variables, including age, employment type, education, occupation, working hours, capital gains/losses, and income category.

Source: UCI Machine Learning Repository — Adult dataset.

4. Data Quality Summary

Number of records: 48,842

Number of columns: 20

Exact duplicate rows retained: 29

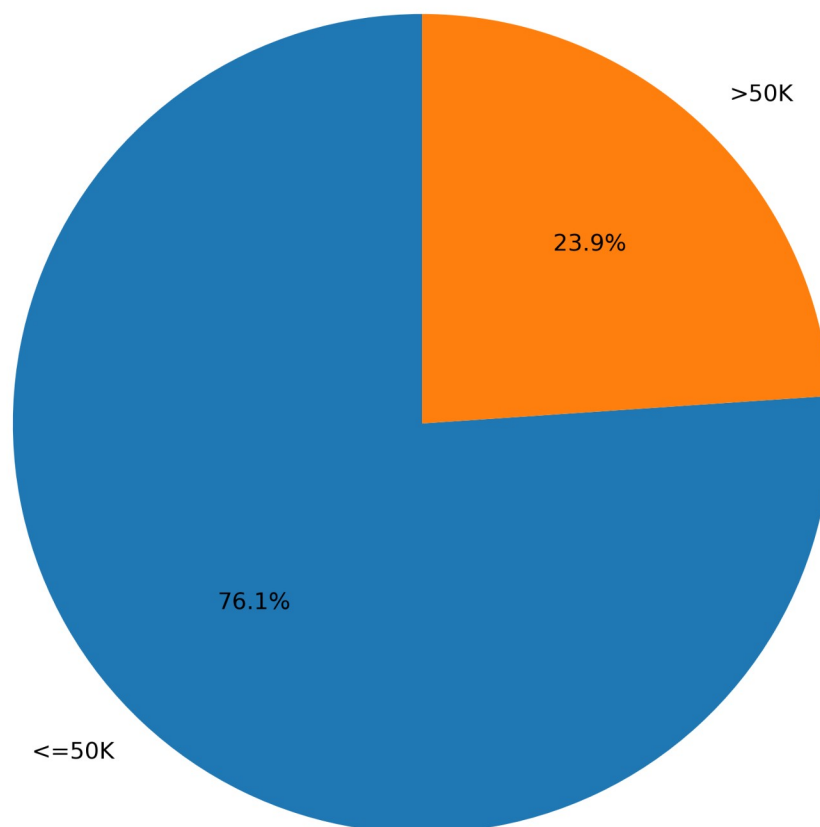
Total remaining missing values: 0

5. Income Distribution

$\leq 50K$: 37,155 records (76.07%)

$> 50K$: 11,687 records (23.93%)

Income Category Proportion



6. Education and Income

The table below summarizes the percentage of records above USD 50K for each education category.

Prof-school: 73.98% above USD 50K

Doctorate: 72.56% above USD 50K

Masters: 54.91% above USD 50K

Bachelors: 41.28% above USD 50K

Assoc-acdm: 25.80% above USD 50K

Assoc-voc: 25.33% above USD 50K

Some-college: 18.96% above USD 50K

HS-grad: 15.86% above USD 50K

12th: 7.31% above USD 50K

7th-8th: 6.49% above USD 50K

10th: 6.26% above USD 50K

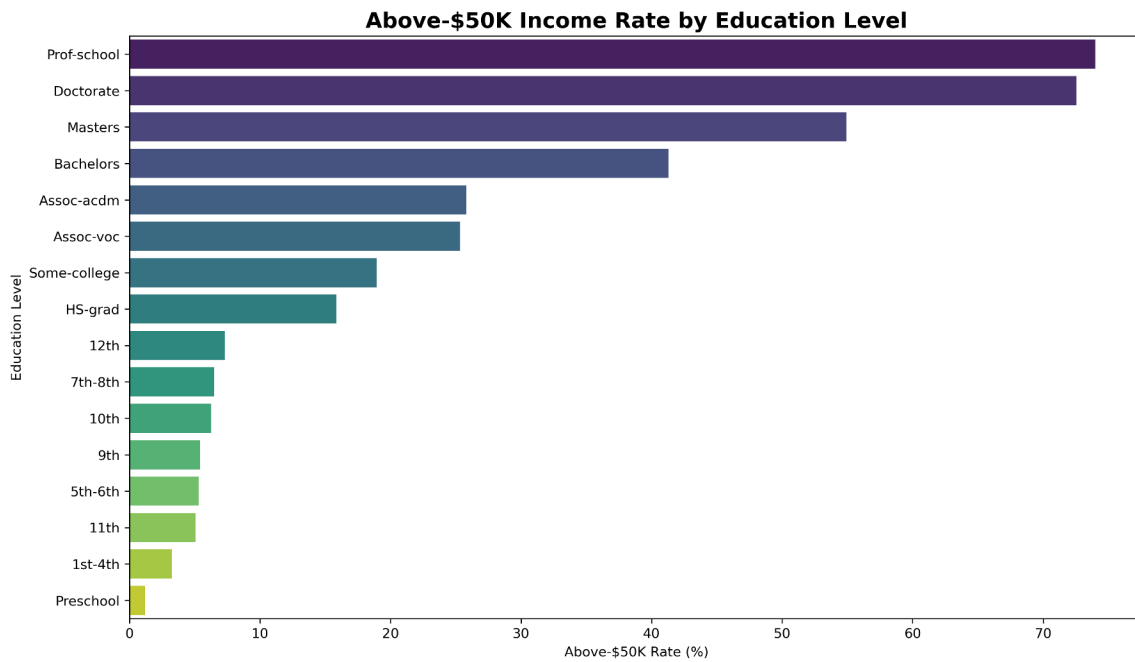
9th: 5.42% above USD 50K

5th-6th: 5.30% above USD 50K

11th: 5.08% above USD 50K

1st-4th: 3.24% above USD 50K

Preschool: 1.20% above USD 50K



7. Age Group and Income

25-34: 16.82% above USD 50K

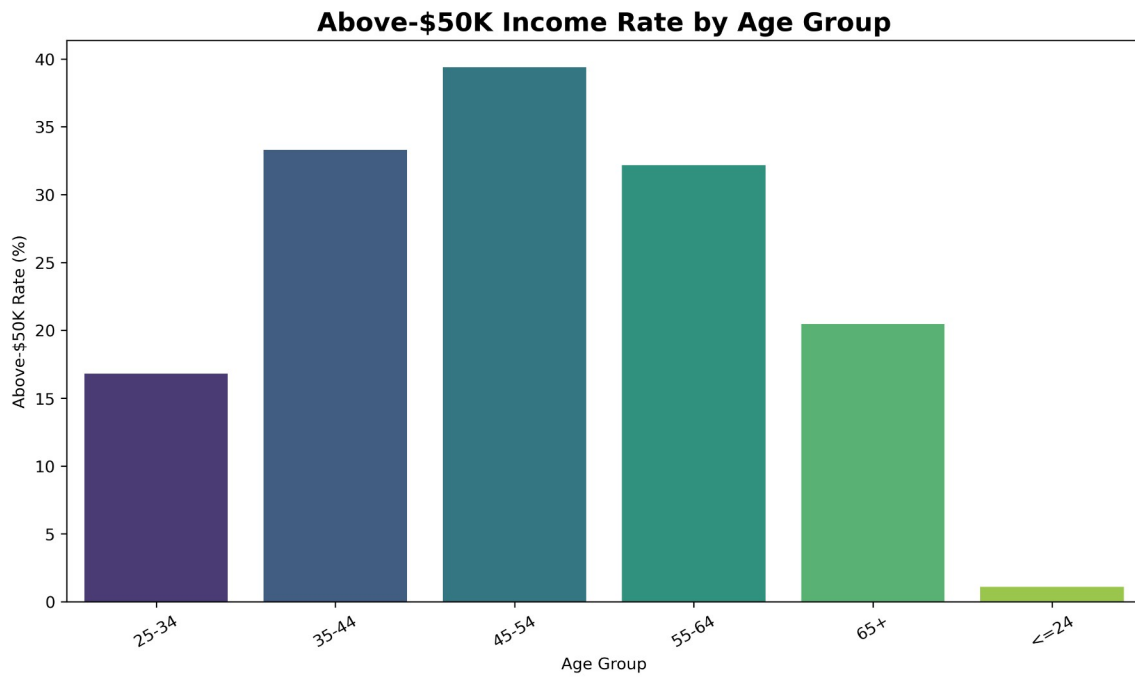
35-44: 33.29% above USD 50K

45-54: 39.39% above USD 50K

55-64: 32.16% above USD 50K

65+: 20.46% above USD 50K

<=24: 1.10% above USD 50K



8. Employment Type and Income

Self-emp-inc: 55.34% above USD 50K

Federal-gov: 39.18% above USD 50K

Local-gov: 29.56% above USD 50K

Self-emp-not-inc: 27.89% above USD 50K

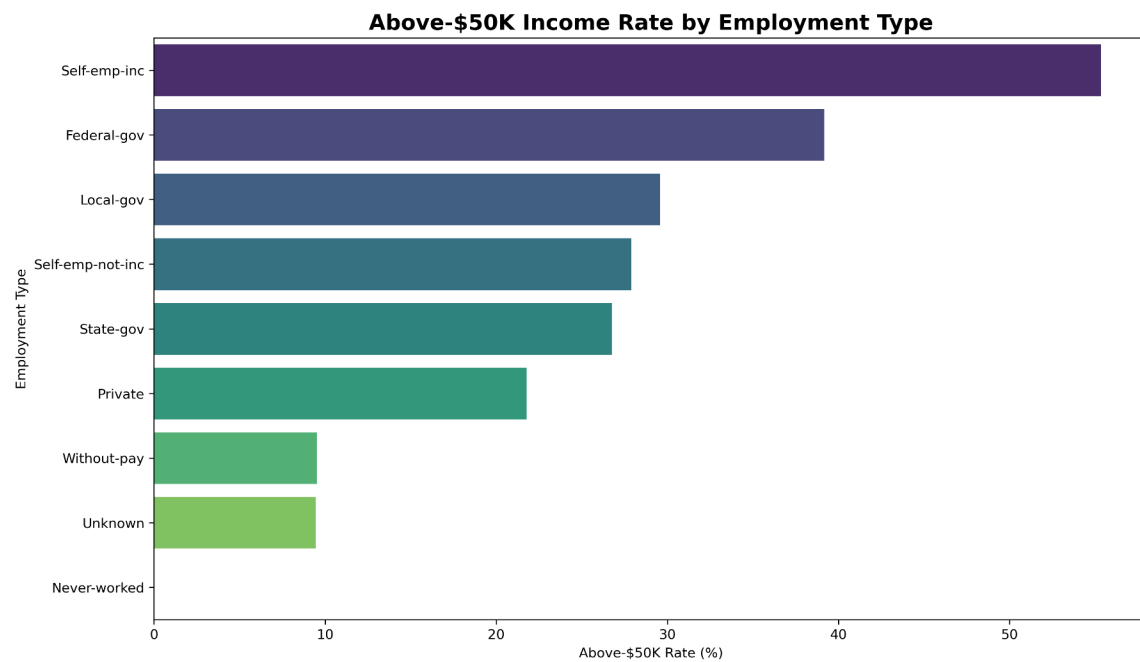
State-gov: 26.75% above USD 50K

Private: 21.79% above USD 50K

Without-pay: 9.52% above USD 50K

Unknown: 9.47% above USD 50K

Never-worked: 0.00% above USD 50K



9. Occupation and Income

Exec-managerial: 47.78% above USD 50K

Prof-specialty: 45.11% above USD 50K

Armed-Forces: 33.33% above USD 50K

Protective-serv: 31.33% above USD 50K

Tech-support: 29.05% above USD 50K

Sales: 26.80% above USD 50K

Craft-repair: 22.63% above USD 50K

Transport-moving: 20.42% above USD 50K

Adm-clerical: 13.69% above USD 50K

Machine-op-inspct: 12.31% above USD 50K

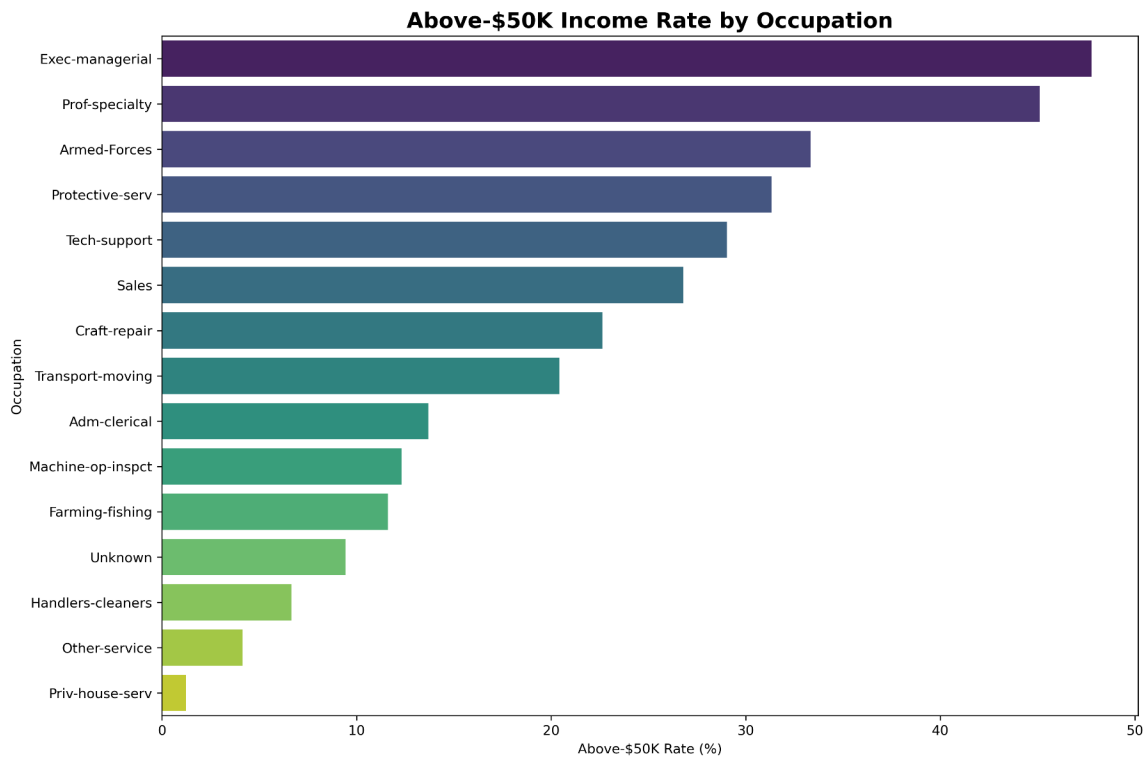
Farming-fishing: 11.61% above USD 50K

Unknown: 9.43% above USD 50K

Handlers-cleaners: 6.66% above USD 50K

Other-service: 4.14% above USD 50K

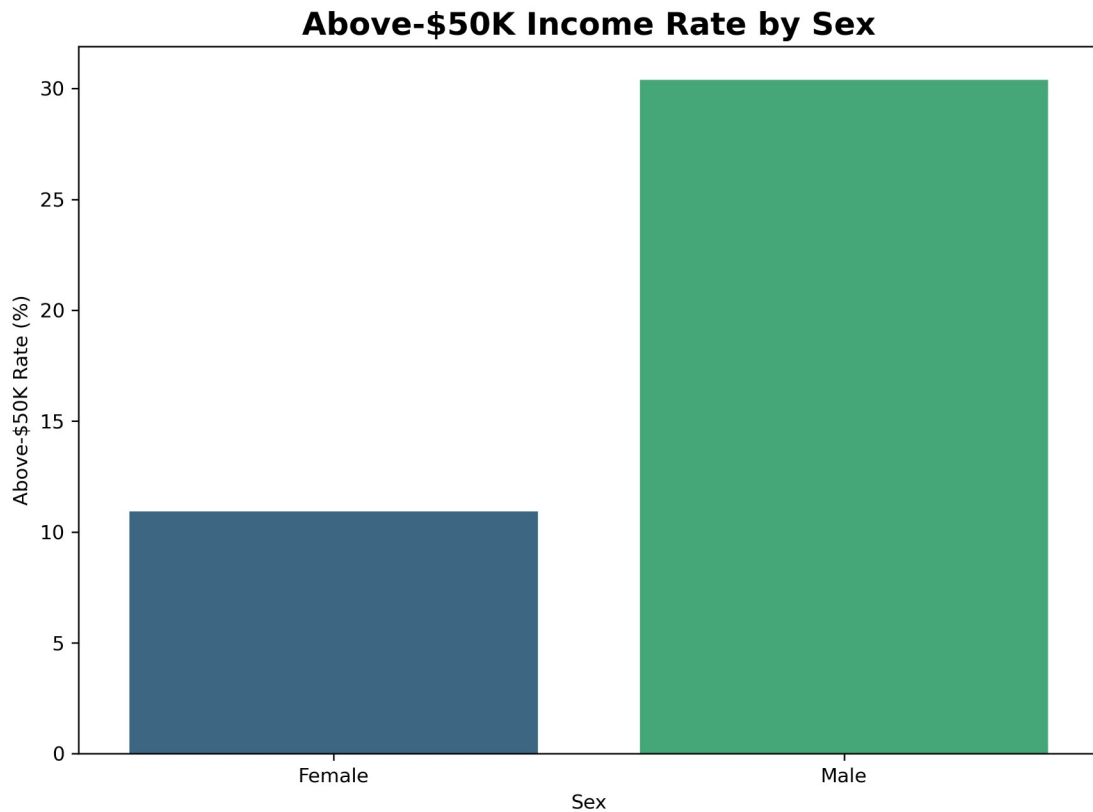
Priv-house-serv: 1.24% above USD 50K



10. Sex and Income

Female: 10.93% above USD 50K

Male: 30.38% above USD 50K



11. Weekly Working Hours and Income

21-34 hours: 7.08% above USD 50K

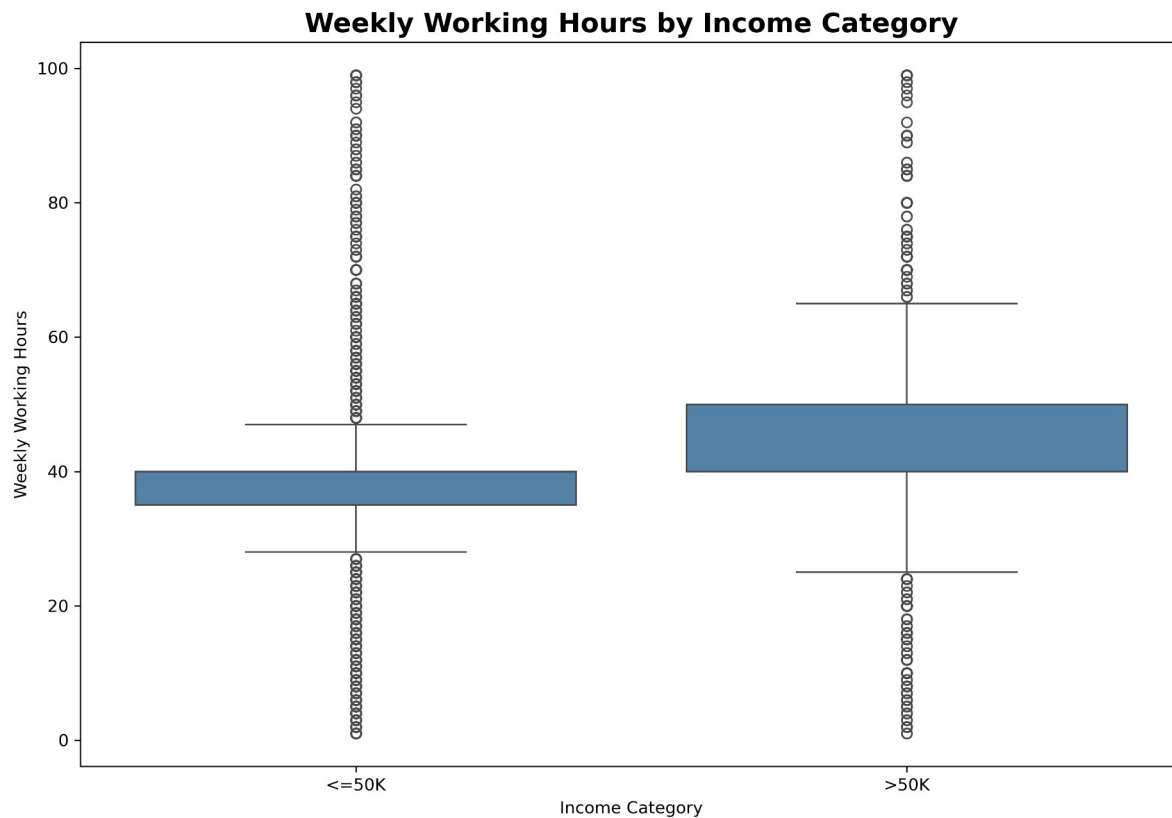
35-40 hours: 20.56% above USD 50K

41-49 hours: 34.64% above USD 50K

50-59 hours: 44.11% above USD 50K

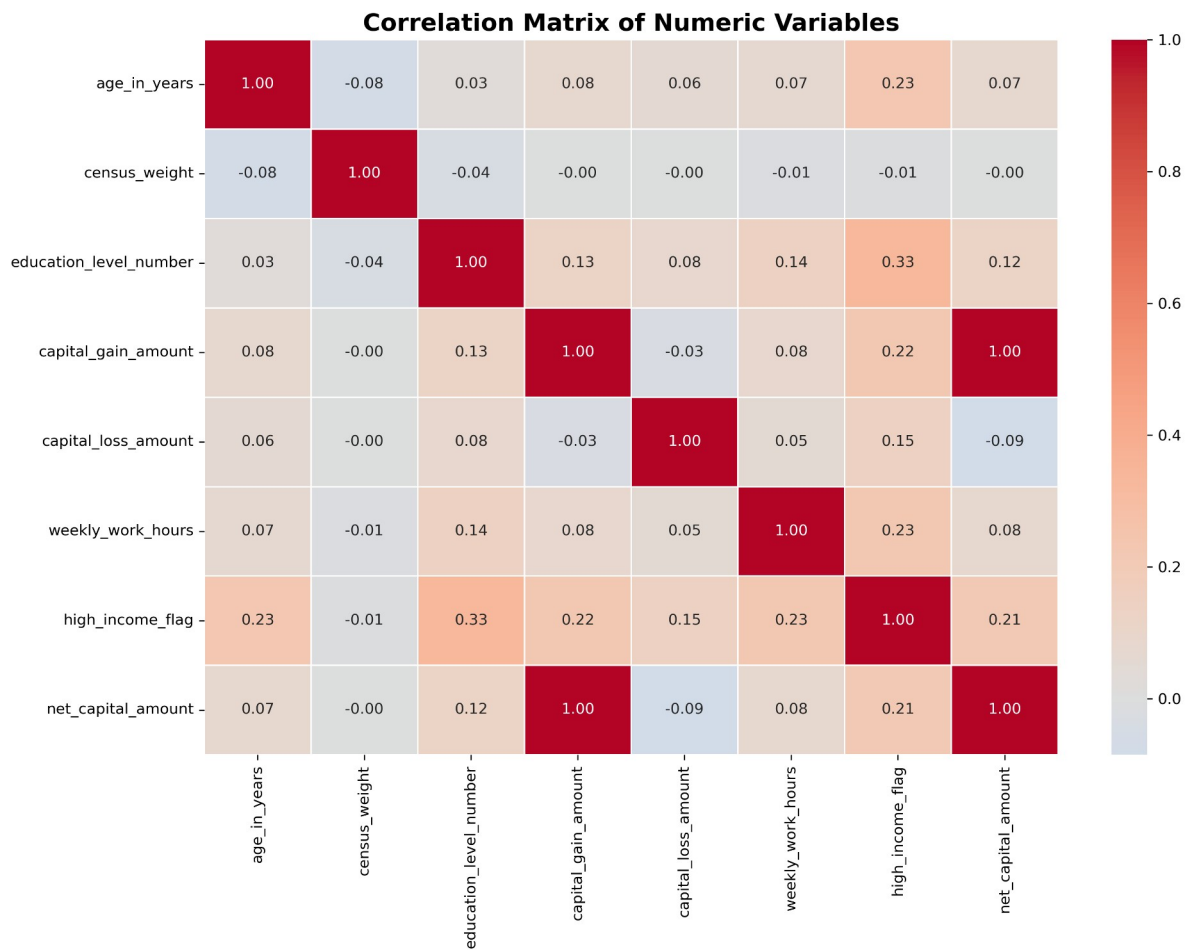
60+ hours: 40.31% above USD 50K

<=20 hours: 6.76% above USD 50K



12. Numeric Relationships

A correlation matrix was examined for selected numerical variables. Correlation measures the strength of a linear association and does not establish causation.



13. Limitations

- The dataset is historical and may not represent current income patterns.
- The analysis is observational and identifies associations rather than causation.
- The dataset contains demographic variables that require careful interpretation.
- The analysis uses descriptive summaries and does not establish predictive performance.
- The census weight variable is not used as a survey-weighted adjustment in the visual summaries.
- Missing categorical values are represented as Unknown rather than being deleted.

14. Conclusion

The exploratory analysis provides a descriptive view of how income categories vary across education, occupation, age groups, employment types, sex, and weekly working hours in the UCI Adult dataset. These patterns can be used as a foundation for further statistical analysis or predictive modeling.